10.1.2 . Reverse a String

A] Algori
thm Step
1: Start
Step 2: Read input string from the user and store it in user_input
Step 3: Define a function reverse_string(input_string)
Step 4: Inside the function, initialize an empty string re = ""
Step 5: Find the length l of input_string
Step 6: Loop from index l-1 down to 0 (reverse order):
    Step 6a: Append the character at current index to re
Step 7: After the loop, return re (the reversed string)
Step 8: Call reverse_string(user_input) and store the result in result
Step 9: Print the original string
user_input **Step 10:** Print the reversed
string result **Step 11:** End

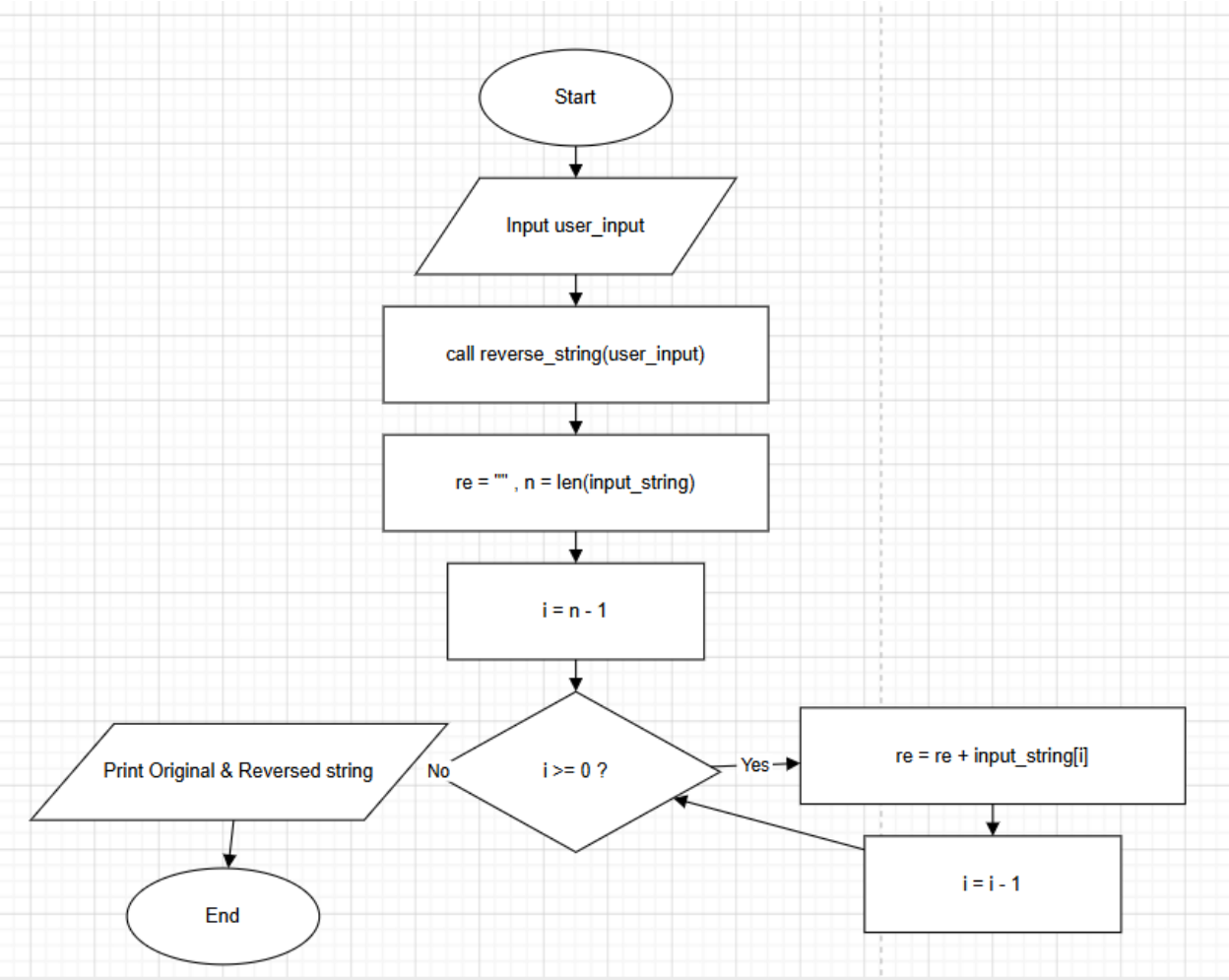
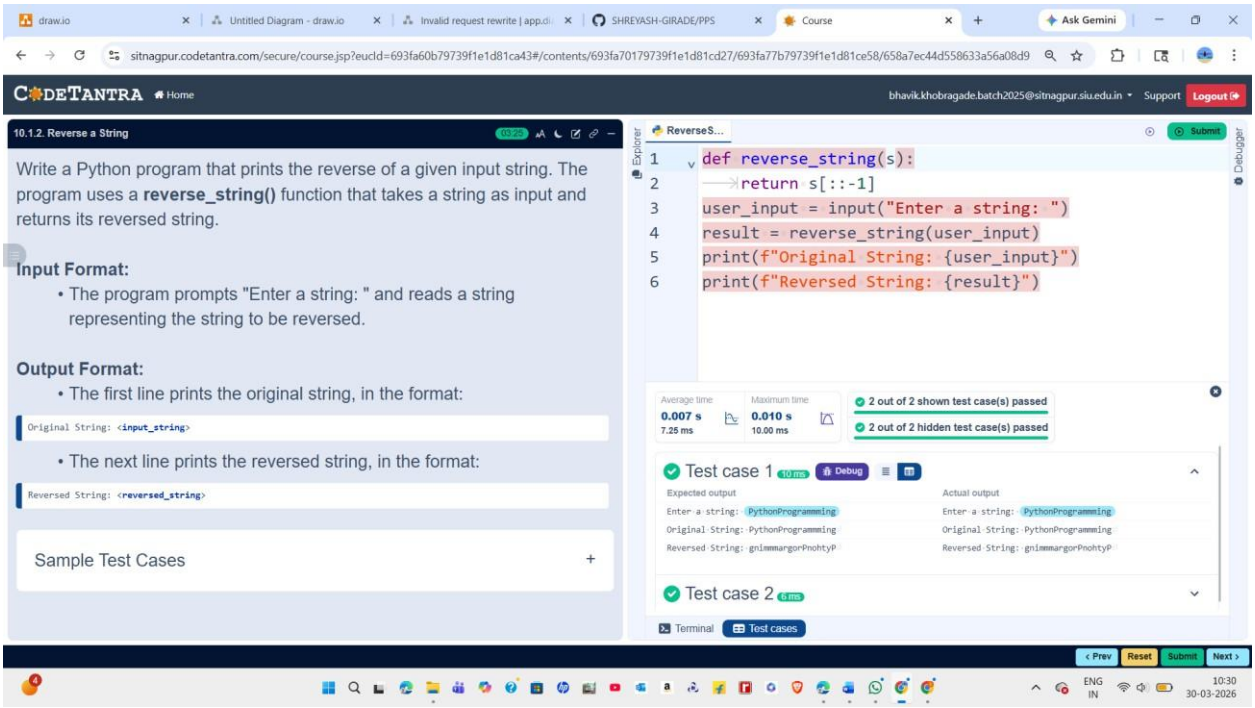
B] code

```
def
        reverse_string(input
        _string): re=""
        l = len(user_input)
        for i in range(l-1,-1,-1):
                re += user_input[i]
        return re



user_input = input("Enter a string: ")
result = reverse_string(user_input)
print(f"Original String: {user_input}")
print(f"Reversed String: {result}")
```

C] flowchart



D] OUTPUT